

Two Incomparable Denominator Exclusions for $\sum_{n \geq 2} (n! - 1)^{-1}$

Finite certificates and exact rationality criteria for Erdős Problem #68

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31 August 2026; revised 18 September 2026

Problem 0.1 (Erdős #68). Is

$$S = \sum_{n \geq 2} \frac{1}{n! - 1}$$

irrational?

ABSTRACT

Any expression of $S = \sum_{n \geq 2} (n! - 1)^{-1}$ as a/q , with integers a and $q > 0$, must satisfy

$$q \nmid 2999999!, \quad q \geq 2^{39990} > 10^{12038},$$

according to the exact finite calculations described here. The first exclusion uses the next integer above a factorial-scaled partial sum; the second uses a rational enclosure and continued fractions. Neither restriction implies the other.

A separate coefficient construction gives remainders of the form $MS + k$, with integers M, k . We classify the vectors that cancel prescribed initial weighted sums and determine their possible M . At fixed M , all remainders have the same fractional part. To prove irrationality by this approach, each positive integer must divide a moment whose remainder is shown to lie strictly between consecutive integers.

1 The denominator exclusions

The factorial-divisibility exclusion. For $m \geq 2$, put

$$H_m = \sum_{n=2}^m \frac{1}{n! - 1}, \quad Z_m = \lfloor m! H_m \rfloor + 1,$$

and for $m \geq 3$ define the carry

$$b_m = mZ_{m-1} + 1 - Z_m.$$

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Here Z_m is the least integer strictly greater than $m!H_m$, even when $m!H_m$ is itself an integer; it is not always $\lceil m!H_m \rceil$. Thus $b_m = 1$ means that the next integers scale exactly by the factor m : $Z_m = mZ_{m-1}$. The exclusion rests on the implication, for integers a and $q > 0$,

$$S = a/q, \quad q \mid (m-1)!, \quad m \geq 3 \implies b_m = 1. \quad (1)$$

For $n \geq 3$, we have $1/(n! - 1) < (n-1)/n!$ and $\sum_{n>m} (n-1)/n! = 1/m!$, so $0 < m!(S - H_m) < 1$. Under the divisibility hypothesis, $m!S$ is integral, so this bound identifies it as Z_m . The same argument at $m-1$ gives $Z_m = m!S = mZ_{m-1}$, proving (1). Thus $b_m \neq 1$ forces every rational denominator to be at least m , since any smaller positive q would divide $(m-1)!$. The recorded exact value $b_{300000} \neq 1$ therefore excludes every divisor of $299999!$ as a denominator of S . This does not exclude every 299999-smooth denominator: sufficiently high powers of a small prime need not divide $299999!$. Section 7 describes the computation of b_{300000} , which is carried out outside Lean.

The size exclusion. For the continued-fraction calculation, put $B = 2^{80000}$ and let N be the first integer with $N! - 1 > B$. Define

$$\ell = \sum_{n=2}^{N-1} \left\lfloor \frac{B}{n! - 1} \right\rfloor, \quad u = \ell + (N-2) + \left\lfloor \frac{2B}{N! - 1} \right\rfloor + 1.$$

Rounding the $N-2$ prefix terms down loses less than $N-2$. Successive terms in the omitted tail have ratio less than $1/(N+1)$, so their sum is less than $(N+1)/(N(N!-1)) < 2/(N!-1)$. Thus $\ell/B < S < u/B$. Exact integer computation gives $N = 7054$ and $u - \ell = 7053$.

Apply the continued-fraction algorithm to both endpoints. Retain a common integer part only while both remainders are nonzero, then invert and reverse the endpoint order. The computation gives 23449 common partial quotients $a_0, \dots, a_{23448}$. Their convergent denominators satisfy

$$Q_{-2} = 1, \quad Q_{-1} = 0, \quad Q_j = a_j Q_{j-1} + Q_{j-2}, \quad Q_{23448} \geq 2^{39990}.$$

Every rational in the open enclosure has this initial segment. To see why it gives a denominator bound, write P_j/Q_j for the last common convergent, with $j = 23448$. A continuation with complete quotient $x/y > 1$, where x, y are coprime positive integers, has reduced denominator $Q_j x + Q_{j-1} y \geq Q_j$; if the expansion terminates at this convergent, its denominator is Q_j . Reducedness follows from the determinant identity $P_j Q_{j-1} - P_{j-1} Q_j = \pm 1$ [12, §1.12(ii), (1.12.5)–(1.12.7), (1.12.20)–(1.12.21)]. The integer comparison $2^{39990} > 10^{12038}$ completes the second exclusion. A non-reduced representation has a denominator at least as large as the reduced one, so the bound applies to every q in $S = a/q$. The long paper, *Denominators and Rationality Criteria*, links the program and its output in §6.

Neither restriction implies the other: the prime 300007 satisfies the divisibility restriction and fails the size restriction, whereas 299999! does the reverse. Both conclusions are finite; they do not decide the irrationality question posed by Erdős [1, p. 102] and listed in Bloom's catalogue [2]. Their computational certificates are specified in §7.

The corresponding rationality criterion. A finite non-unit carry excludes particular denominators. Non-unit carries at arbitrarily large indices are equivalent to irrationality.

Theorem 1.1 (exact carry characterisation). *The following conditions are equivalent:*

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$$S \notin \mathbb{Q}, \quad (\forall B)(\exists m > B) b_m \neq 1, \quad (\forall B)(\exists m > B) m \nmid Z_m.$$

In particular, *cofinal non-unit carries imply irrationality*; equivalently, the original problem is the criterion using the next integer.

Proof. Rationality forces eventual unit carries by (1). Conversely, if $b_m = 1$ eventually, then $Z_m/m!$ is eventually constant. Since

$$H_m < Z_m/m! \leq H_m + 1/m!,$$

that rational constant is S . Finally writing $x = (m-1)!H_{m-1}$ gives $b_m = m-1 - \lfloor m\{x\} + 1/(m!-1) \rfloor$, hence $-1 \leq b_m \leq m-1$. For $m \geq 3$, these bounds together with $Z_m = mZ_{m-1} + 1 - b_m$ imply $m \mid Z_m$ exactly when $b_m = 1$. $\square$

For e , multiplication by $m!$ clears every denominator of the partial sum; here $\gcd(m!, m!-1) = 1$ prevents even the last summand from being cleared when $m \geq 3$. For the tail arguments below, the issue is the gap to an integer after scaling, not just the size of the unscaled tail. The remaining sections study that gap and the exact coefficient equations. Appendix A gives a factorial-digit formulation and the floor correction relating the two sequences. Secondary constructions are proved in the long paper.

Formal proofs. A result with a kernel-checked Lean proof carries a mark in the margin. Lean opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. Comparator opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean's Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The evidence record gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

2 Integer vectors for cancelling weighted sums

For a chosen integer $D \geq 2$, we seek integer coefficients whose contributions at denominators $2!-1, \dots, D!-1$ vanish, removing the first few terms of a remainder. To construct such coefficients, we remove powers of $d!$ from $n!$. For example, at $d = 3$ and $n = 4$, removing the factor $3! = 6$ leaves 4, and $24/5 - 4/5 = 4$: the fractional part is unchanged. For a finitely supported integer vector $\lambda = (\lambda_n)_{n \geq 1}$ and an integer $d \geq 2$, define

$$W_{d,n} = \frac{n!}{(d!)^{\lfloor n/d \rfloor}}, \quad M(\lambda) = \sum_n \lambda_n n!, \quad V_d(\lambda) = \sum_n \lambda_n W_{d,n}.$$

The **weights are integers**. Indeed, write $n = kd + r$ with $0 \leq r < d$. The quotient $n!/((d!)^k r!)$ is a multinomial coefficient, and $W_{d,n}$ is $r!$ times that integer. Since $n! = (d!)^{\lfloor n/d \rfloor} W_{d,n}$ and $d! \equiv 1 \pmod{d! - 1}$, the difference $(n! - W_{d,n})/(d! - 1)$ is an integer. Thus the same fractional-part calculation holds for every $d \geq 2$ and $n \geq 1$. Write e_n for the vector with coefficient 1 at index n and 0 elsewhere. We temporarily allow index 1 to simplify the basis calculation; we later require its coefficient to vanish. This does not add the undefined term $1/(1! - 1)$ to S .

The exponent $\lfloor n/d \rfloor$ is chosen for its changes at multiples of d , not to remove the largest power of $d!$ dividing $n!$. Consequently, the adjacent differences below affect only sums with $d \mid n$. Call $M(\lambda)$ the moment, without assuming it positive. The remainder

$$\mathcal{R}(\lambda) = \sum_{d \geq 2} \frac{V_d(\lambda)}{d! - 1}$$

converges absolutely: beyond the support, $V_d = M$. For any $N \geq 2$ at least as large as every supported index, the congruences just proved give

$$\mathcal{R}(\lambda) - M(\lambda)S = \sum_{d=2}^N \frac{V_d(\lambda) - M(\lambda)}{d! - 1} \in \mathbb{Z}.$$

Thus the weights retain an integer linear form in 1 and S while allowing $V_2, \dots, V_D$ to be set to zero. We now solve those equations.

For example, $4e_3 - e_4$ has moment zero, $V_2 = 6$ and $V_4 = 23$, with all other V_d zero. The vector $2e_1 - e_2$ has moment zero and only $V_2 = 1$ nonzero. Subtracting six copies of the latter therefore removes the unwanted V_2 without changing V_4 . The following recursion performs this elimination at the proper divisors of each index.

Theorem 2.1 (an integer basis with prescribed weighted sums). *Set*

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$$T_n = ne_{n-1} - e_n, \quad U_n = T_n - \sum_{\substack{d \mid n \\ 2 \leq d < n}} W_{d,n} U_d \quad (n \geq 2).$$

Then

$$M(U_n) = 0, \quad V_d(U_n) = (d! - 1)\mathbf{1}_{d=n}.$$

The vectors $e_1, U_2, U_3, \dots$ form an integral basis. Every finite vector has the unique finite expansion

$$\lambda = M(\lambda)e_1 + \sum_{d \geq 2} \frac{V_d(\lambda) - M(\lambda)}{d! - 1} U_d. \quad (2)$$

Proof. The moment of T_n is zero. The floor in $W_{d,n}$ changes between $n-1$ and n precisely when $d \mid n$. Accordingly, **the d th weighted sum of T_n vanishes unless d divides n** , and in full

$$V_d(T_n) = (d! - 1)W_{d,n}\mathbf{1}_{d \mid n}.$$

The recursion cancels the weighted sums indexed by proper divisors, proving the two identities by induction. The columns $e_1, T_2, \dots, T_N$ form a triangular integer matrix with diagonal entries $1, -1, \dots, -1$. Its determinant is ± 1 , so it is unimodular and its columns

form a $\mathbb{Z}$ -basis. The change from T_n to U_n is also triangular over $\mathbb{Z}$, with diagonal entries 1. Thus $e_1, U_2, \dots, U_N$ is a $\mathbb{Z}$ -basis as well. Applying M and each V_d identifies its coefficients as in (2). For d beyond the support, $V_d = M$, so no infinite sum is required. $\square$

For a prime $p \geq 3$, the recursion gives $U_p = T_p$: adding U_p changes V_p by $p! - 1$, leaves every other V_d unchanged, and leaves M unchanged. The same construction works at composite indices. For example,

$$U_9 = 9e_8 - e_9 - 5040e_2 + 1680e_3. \quad (3)$$

It has moment zero and only V_9 is nonzero. These individual adjustments need not preserve support on $n \geq 2$: the earlier U_4 has coefficient -12 at index 1. We now impose that restriction.

2.1 The support restriction and the remainder

Fix $D \geq 2$ and write

$$L_D = \text{lcm}_{2 \leq d \leq D}(d! - 1), \quad K_D = L_D e_1 - \sum_{d=2}^D \frac{L_D}{d! - 1} U_d.$$

The coefficients in (2) are integers, so

$$V_d(\lambda) \equiv M(\lambda) \pmod{d! - 1}. \quad (4)$$

When $V_2 = \dots = V_D = 0$, these congruences force $L_D \mid M$. Substituting $M = tL_D$ gives every such solution, with finitely many nonzero z_n :

$$V_2 = \dots = V_D = 0 \iff \lambda = tK_D + \sum_{n>D} z_n U_n, \quad M = tL_D. \quad (5)$$

The basis identities give $\mathcal{R}(e_1) = S$ and $\mathcal{R}(U_n) = 1$. By linearity,

$$\mathcal{R}\left(tK_D + \sum_{n>D} z_n U_n\right) = tL_D(S - H_D) + \sum_{n>D} z_n. \quad (6)$$

The finite integer-difference identity shows that changing coefficients at fixed moment preserves the fractional part. A nonintegral remainder rules out $S = a/q$ when $q \nmid M$. To exclude every denominator this way, each positive integer q must divide the moment of some nonintegral remainder in the family; unbounded moments alone do not guarantee this.

Let a_D and u_n be the coefficients at index 1 of K_D and U_n , respectively. Requiring that coefficient to vanish gives

$$ta_D + \sum_{n>D} z_n u_n = 0. \quad (7)$$

Consequently the attainable moments are

$$L_D \frac{g_D}{\gcd(g_D, a_D)} \mathbb{Z}, \quad g_D = \gcd\{u_n : n > D\}. \quad (8)$$

Indeed, finite integer combinations of the u_n form $g_D \mathbb{Z}$. The positive generator is attained because a gcd of any integer family is a finite integer combination of its members. An attaining vector is primitive: division by a common coefficient factor greater than one would preserve all zero weighted sums and produce a smaller positive moment. For odd n , $u_n = 0$ by induction. For a prime p , the sole nonzero proper-divisor contribution at $2p$ gives $u_{2p} = -2(2p)!/2^p \neq 0$. Every tail thus contains a nonzero coefficient, so $g_D > 0$.

Computing the gcd from finitely many coefficients. The scalar recurrence is

$$u_2 = 2, \quad u_n = - \sum_{\substack{d|n \\ 2 \leq d < n}} W_{d,n} u_d \quad (n > 2).$$

To prove that an integer g divides every u_n with $n > D$, use induction in this recurrence. Once the terms with $D < d < n$ are divisible by g , only $d \leq D$ need further consideration. For these terms we use

$$k! \mid W_{d,dk} = \frac{(dk)!}{(d!)^k}.$$

The quotient by $k!$ counts partitions into k unordered blocks of size d .

Theorem 2.2 (a finite formula for the gcd). *Choose a prime ℓ with $D/2 < \ell \leq D$ and put $H = D(2\ell - 1)$. Then*

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$$g_D = \gcd(u_{D+1}, \dots, u_H), \quad H < 2D^2.$$

Proof. Since $D < 2\ell \leq D(2\ell - 1) = H$, the finite interval includes the nonzero term $u_{2\ell} = -(2\ell)!/2^{\ell-1}$. Thus its gcd g is positive and $g \mid (2\ell)!$. For $n > H$ and a divisor $d \leq D$, the integer n/d is at least 2ℓ . The equal-block divisibility above gives $g \mid (n/d)! \mid W_{d,n}$. Every term of the recurrence with $d \leq D$ is therefore divisible by g . Strong induction handles the remaining divisors $D < d < n$, whose coefficients are already divisible by g . Thus g divides the entire tail. Finally $H \leq D(2D - 1) < 2D^2$.

This does not justify replacing H by $2D$. □

Bertrand's postulate supplies ℓ for every $D \geq 3$, and $\ell = 2$ works for $D = 2$. This finite calculation determines allowable moments, not the nonintegrality of their remainders.

Theorem 2.3 (constant values of the floor in the weights). *Let λ be a finitely supported integer vector, let $d \geq 2$ and $k \geq 0$ be integers, and suppose each index n in its support satisfies $kd \leq n < (k+1)d$. Then*

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$$M(\lambda) = (d!)^k V_d(\lambda).$$

In particular, support in $[d, 2d)$ and $V_d(\lambda) = 0$ force $M(\lambda) = 0$. If all supported indices are at least d , $V_d(\lambda) = 0$ and $M(\lambda) \neq 0$, some supported index is at least $2d$.

Proof. On the stated interval $\lfloor n/d \rfloor = k$, so $n! = (d!)^k W_{d,n}$. Sum against λ_n ; the final assertion follows by contraposition from the case $k = 1$. □

For example, at $D = 4$ the minimum moment is 1380, attained by $12K_4 + 253U_6 - 11U_8$. The inputs to (8) are $L_4 = 115$, $a_4 = -55$ and $g_4 = 60$. The last equality uses the full-tail divisibility $\text{lcm}(1, \dots, n) \mid u_n$ together with $23u_6 - u_8 = 60$. The long paper, §5.1, proves that divisibility by Legendre's formula, checks the attaining vector, and gives the different minimum 4140 when support is restricted to $n \leq 6$.

There is also a short certificate for this depth-four value that does not require computing the entire tail gcd. For every $n \geq 2$,

$$1380 \mid 11n! - 46W_{2,n} + 12W_{4,n}. \quad (9)$$

For $n = 2, 3, 4$ the expression is zero. For $n \geq 4$, each summand is divisible by 12: $12 \mid n!$, while $6 \mid W_{2,n}$ because $W_{2,2r} = r! \prod_{j=1}^r (2j-1)$ for $r \geq 2$ and $W_{2,2r+1} = (2r+1)W_{2,2r}$. For $n \geq 5$, each summand is divisible by 5, since the denominators removed from $n!$ in $W_{2,n}$ and $W_{4,n}$ are powers of 2 and 24. Finally, $n! = 24^{\lfloor n/4 \rfloor} W_{4,n} \equiv W_{4,n} \pmod{23}$ and $46W_{2,n} \equiv 0 \pmod{23}$. The factors 12, 5 and 23 are coprime, proving (9).

Summing against a finite integer vector gives $1380 \mid 11M - 46V_2 + 12V_4$. Thus $V_2 = V_4 = 0$ implies $1380 \mid M$, since 11 is coprime to 1380. Equality is attained even with the third channel cancelled:

$$\lambda = 1482e_2 - 784e_3 - 136e_5 + 83e_6 - e_8, \quad (M, V_2, V_3, V_4) = (1380, 0, 0, 0). \quad (10)$$

These values follow directly from the five rows $(n!, W_{2,n}, W_{3,n}, W_{4,n})$ at $n = 2, 3, 5, 6, 8$: $(2, 1, 2, 2)$, $(6, 3, 1, 6)$, $(120, 30, 20, 5)$, $(720, 90, 20, 30)$ and $(40320, 2520, 1120, 70)$. Hence imposing $V_3 = 0$ does not change the attainable moments; it does restrict the vectors, since $3e_2 - e_3$ has $M = V_2 = V_4 = 0$ but $V_3 = 5$.

The first possible maximum index of a vector attaining moment 1380 under $V_2 = V_4 = 0$ is 8. Indeed, the expression in (9) is zero at $n = 2, 3, 4, 5$ and equals 4140 and $28980 = 7 \cdot 4140$ at $n = 6, 7$. For support at most 7, $11M = 4140(\lambda_6 + 7\lambda_7)$ when the two channels vanish; coprimality of 11 and 4140 forces $4140 \mid M$. The vector in (10) reaches index 8. All of these statements concern finite cancellation; they do not prove irrationality of the series.

3 How cancellation forces a large moment

On the actual support $n \geq 2$, one has the stronger congruence $12 \mid M - 2V_2$. For $n = 2, 3$, $n! = 2W_{2,n}$. For $n \geq 4$, both $n!$ and $2W_{2,n}$ are divisible by 12: writing $n = 2k$ or $2k+1$, $W_{2,2k} = k! \prod_{j=1}^k (2j-1)$ is divisible by 6 for $k \geq 2$. Since every $d! - 1$ is coprime to 6, vanishing of $V_2, \dots, V_D$ implies

$$12L_D \mid M. \quad (11)$$

This is a necessary divisor, not the minimum: the long paper, §5.1, gives examples attaining $12L_D$ at $D = 2, 3$ but proves that the minimum at $D = 6$ is $24L_6$. It does not by itself cover every fixed denominator either: L_D is coprime to 3, so $9 \nmid 12L_D$ for every D .

3.1 Growth of the compulsory factor

The divisibility $L_D \mid M$ forces any nonzero moment with $V_2 = \dots = V_D = 0$ to be large. To estimate it, take a block of denominators near N : their product is large, and pairwise

gcds bound how much can be lost on passing to the lcm. This gives

$$\liminf_{N \rightarrow \infty} \frac{\log L_N}{N^{3/2} \log N} \geq \frac{2\sqrt{2}}{3}. \quad (12)$$

This bounds a common denominator before reduction, not a denominator of S . Multiplying $i! - 1$ by $j!/i!$ and subtracting $j! - 1$ gives $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$. For positive integers x_i , the divisibility $\prod_i x_i \mid \text{lcm}(x_i) \prod_{i < j} \gcd(x_i, x_j)$ follows prime by prime: after ordering valuations, each except the maximum occurs at least once among the pairwise minima. On a terminal block of k terms, this inequality and $\gcd(i! - 1, j! - 1) \mid j!/i! - 1$ give

$$\log L_N \geq \sum_{n=N-k+1}^N \log(n! - 1) - \binom{k+1}{3} \log N.$$

The product contributes about $kN \log N$, whereas the gcd loss is of order $k^3 \log N$. Balancing them suggests k of order $\sqrt{N}$. Fix $\alpha > 0$ and take $k = \lfloor \alpha \sqrt{N} \rfloor$. Dividing the inequality by $N^{3/2} \log N$ and letting $N \rightarrow \infty$ gives $\liminf \log L_N / (N^{3/2} \log N) \geq \alpha - \alpha^3/6$. The choice $\alpha = \sqrt{2}$ proves (12); the error estimates are given in the long paper, §2.

This lower bound does not provide a small positive scaled tail. For $N \geq 3$, any common divisor of $(N-1)! - 1$ and $N! - 1$ divides $N-1$ by subtraction, and is coprime to $N-1$ because $N-1 \mid (N-1)!$. Thus the two denominators are coprime. Since $S - H_N > 1/((N+1)! - 1)$,

$$L_N(S - H_N) > \frac{((N-1)! - 1)(N! - 1)}{(N+1)! - 1} \rightarrow \infty.$$

Even two terms therefore obstruct clearing by the full common denominator; (12) gives the stronger quantitative cost.

3.2 Solutions supported on an arithmetic progression

For $D, r \geq 2$ and a positive multiple ℓ of $2, \dots, D$, there is a primitive integer vector supported on $r, r + \ell, \dots, r + (D-1)\ell$. With $N = r + (D-1)\ell$, it satisfies

$$V_2 = \dots = V_D = 0, \quad M = N! \prod_{d=2}^D (1 - (d!)^{-\ell/d}) > 0, \quad (\lfloor \sqrt{N}/2 \rfloor + 1)! \mid M.$$

The long paper, §5, paragraph “A primitive solution on an arithmetic progression”, constructs the coefficients from a polynomial with prescribed roots and proves integrality and primitivity by a multinomial count. Thus $N \geq 4(q-1)^2$ guarantees $q \mid M$ for each positive integer q , including for denominators increasing with N . What remains unproved is nonintegrality of the corresponding remainders; the divisibility and the successful comparison must concern the same vector.

4 Cancellation in a reduced prefix

A large common denominator can shrink on addition: $1/3 + 1/15 = 2/5$ removes the prime 3, whereas $1/9 + 1/45 = 2/15$ only lowers its exponent. Write $v_q(a)$ for the exponent of the prime q in a positive integer a , and $\text{den}(x)$ for a rational number's positive reduced denominator.

Louwsma and Martino give a valuation formula for an elementary symmetric sum [7, Lemma 4.1, p. 10]. Dividing that sum by the product of its arguments gives the reciprocal-sum test below; we prove it directly for $n! - 1$. Put $L_M = \text{lcm}_{2 \leq n \leq M} (n! - 1)$ and $A_M = \sum_{n=2}^M L_M / (n! - 1)$. For a prime q with attained maximum $e = \max_{2 \leq n \leq M} v_q(n! - 1) > 0$, set

$$I_{q,e}(M) = \{n \in [2, M] : v_q(n! - 1) = e\}, \quad R_{q,e}(M) = \sum_{n \in I_{q,e}(M)} \left(\frac{n! - 1}{q^e} \right)^{-1} \pmod{q}.$$

After multiplication by L_M , all lower-valuation terms vanish modulo q . Hence

$$A_M \equiv (L_M / q^e) R_{q,e}(M) \pmod{q},$$

and L_M / q^e is a unit. Consequently

$$v_q(\text{den } H_M) = e \iff R_{q,e}(M) \not\equiv 0 \pmod{q}. \quad (13)$$

The equivalence, with the maximum and its attainment taken from the actual prefix lcm, is [maximal prime power survival](#).

Reduction of $H_M = A_M / L_M$ gives $v_q(\text{den } H_M) = \max\{0, e - v_q(A_M)\}$. A zero residue in (13) therefore lowers the exponent; complete cancellation requires $v_q(A_M) \geq e$. For $e = 1$ these conditions coincide, but for $e > 1$ they need not.

A unique maximal exponent survives because its residue sum has one nonzero term. For repeated maxima, cancellation does occur in the actual sequence. At $M = 138$, the maximal 139-exponent occurs at 69, 122, 137, with cofactor residues 6, 49, 73. The exact modular scan gives

$$6^{-1} + 49^{-1} + 73^{-1} \equiv 0 \pmod{139}.$$

The scan for 2593 also gives a zero reciprocal sum; its indices and cofactor residues are recorded in the long paper, §1. All attained exponents in these two examples are one, so each prime disappears completely. The cancellations already hold at indices 137 and 2591, respectively, and persist thereafter. For a prime $q \geq 3$, Wilson's theorem gives $(q - 1)! - 1 \equiv -2 \pmod{q}$, and $n! - 1 \equiv -1 \pmod{q}$ for $n \geq q$. Thus no summand after index $q - 2$ can change the prime's exponent in the reduced denominator; the general proof is in the long paper, §1. This concerns partial sums, not a hypothetical denominator of S . Survival still needs to be combined with a bound for the real tail.

5 Comparing the tail with the distance to an integer

We return to the integer-gap argument of §1, now choosing a scale that clears every prime-power level shared by two summand denominators. Each remaining prime then occurs in just one scaled denominator and cannot cancel. This sufficient choice may exceed what cancellation in the sum requires. Including $(p - 1)!$ ensures that the scale eventually absorbs each fixed rational denominator. For an integer parameter $p \geq 3$, not necessarily prime, write $d_n = n! - 1$ and put

$$F_p = (p - 1)!, \quad D_p = \text{lcm}_{2 \leq i < j \leq 2p-1} \gcd(d_i, d_j),$$

$$C_p = \text{lcm}(F_p, D_p), \quad L_p^{\text{blk}} = \text{lcm}(F_p, d_2, \dots, d_{2p-1}),$$

$$R_p = L_p^{\text{blk}} / C_p, \quad T_p = \sum_{n=2}^{2p-1} L_p^{\text{blk}} / d_n, \quad \rho_p = (-T_p) \bmod R_p.$$

For a prime dividing R_p , the largest exponent among the d_n exceeds both the second-largest exponent and the exponent in F_p . It is attained at exactly one index. Reducing T_p modulo that prime leaves only the term at that index, which is a unit. Thus $\gcd(T_p, R_p) = 1$ and $0 < \rho_p < R_p$ whenever $R_p > 1$.

Thus $R_p = \text{den}(C_p H_{2p-1})$: it is the reduced denominator of the scaled prefix, not generally of H_{2p-1} . No coprimality of C_p and R_p is required. For $R_p > 1$, the gap from T_p/R_p to the next integer is ρ_p/R_p . For $R_p = 1$, the prefix is integral and that strict gap is 1, although $\rho_p = 0$.

For $p = 3$, the denominators 1, 5, 23, 119 are pairwise coprime, so $D_3 = 1$ and $C_3 = 2$. Direct calculation gives

$$C_3 H_5 = \frac{34264}{13685}, \quad \frac{\rho_3}{R_3} = \frac{6791}{13685}.$$

The tail bound used below gives $C_3(S - H_5) < 7/1080 < 6791/13685$, so this block satisfies the required comparison. One block is not enough: the argument needs arbitrarily large p to ensure that any fixed rational denominator eventually divides F_p .

Proposition 5.1 (a sufficient tail inequality). *If arbitrarily large natural parameters $p \geq 3$ satisfy*

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$$R_p > 1, \quad (2p+1)L_p^{\text{blk}} < 2p^2(2p-1)!\rho_p, \quad (14)$$

then S is irrational.

Proof. Suppose $S = a/q$ with $q > 0$, and choose a parameter in the hypothesis with $p > q$. Then $q \mid F_p \mid C_p$, so $C_p S$ is an integer strictly above $C_p H_{2p-1}$. Hence

$$C_p(S - H_{2p-1}) \geq \frac{\rho_p}{R_p}.$$

But $1/(n-1) < 2/n!$ for $n \geq 2p$ and the geometric majorant with ratio $1/(2p+1)$ give

$$C_p(S - H_{2p-1}) < \frac{(2p+1)C_p}{2p^2(2p-1)!} < \frac{\rho_p}{R_p},$$

where the last inequality follows by rearranging (14) and using $L_p^{\text{blk}} = C_p R_p$. This is a contradiction. $\square$

Under $R_p > 1$ we have $\rho_p > 0$, so logarithms are defined. Dividing the strict inequality by R_p and by $F_p = (p-1)!$ gives the equivalent logarithmic comparison

$$\log \frac{C_p}{(p-1)!} - \log \frac{\rho_p}{R_p} < \log \frac{2p^2(2p-1)!}{(2p+1)(p-1)!}. \quad (15)$$

The right side is $p \log p + (2 \log 2 - 1)p + O(\log p)$.

The hypothesis compares a scaled positive tail with the actual gap to the next integer. If $\rho_p/R_p \geq 1/2$, the logarithmic cost of that gap is at most $\log 2$. If $\rho_p = 1$, the gap is only $1/R_p$ and the cost is $\log R_p$. Coprimality permits both cases: it proves positivity, not a uniform lower bound. No unbounded family satisfying the displayed comparison is established here. The factor $C_p/(p-1)!$ and the gap ρ_p/R_p must be controlled at the same indices. Selected-prime and joint-mean variants are recorded in the appendix and proved in the long paper, §§4 and 7; neither is needed for this proposition.

6 The remaining real comparison

The exact target remains the [criterion using the next integer](#):

$$(\forall B)(\exists m > B) \ m \nmid Z_m. \quad (16)$$

For a vector λ with $V_2 = \dots = V_D = 0$ and $M > 0$, choose a cutoff $N \geq D$ at least as large as every supported index. Write

$$A_N = \sum_{d=D+1}^N \frac{V_d(\lambda)}{d! - 1}, \quad \mathcal{R}(\lambda) = A_N + M \sum_{d>N} \frac{1}{d! - 1}.$$

The sufficient strict comparison is

$$\frac{2M}{(N+1)! - 1} < \lfloor A_N \rfloor + 1 - A_N. \quad (17)$$

It puts the remainder strictly between A_N and the next integer. The finite integer-difference identity proved in §2 then excludes $S = a/q$ for every $q \mid M$. For the progression vectors of §3.2, every fixed q divides M once their support endpoint is large enough. Their moment formula also gives $0 < M < N!$, so at that endpoint

$$0 < M(S - H_N) < \frac{2M}{(N+1)! - 1} < \frac{2}{N}.$$

Thus the tested upper bound already tends to zero along this family. The missing step is comparison with the gap on the right of (17), which may also tend to zero. Smallness of the omitted tail alone does not prove the strict comparison. The finite signed sum A_N cannot be omitted. An integral A_N , even a negative one, leaves a gap of 1; a value just below an integer leaves a much smaller gap. Positivity is required of M , not of A_N .

The cutoff need not be the largest supported index. For the vector $-6e_2 + e_4$, with $D = 2$ and moment 12, the comparison fails at $N = 4$: its gap is $9/115$, smaller than $24/119$. At $N = 5$, the gap is $13376/13685$, larger than the new bound $24/719$, so the same vector succeeds. This illustrates a choice of truncation, not an improvement of either denominator exclusion.

For a fixed vector, (17) holds at every sufficiently large cutoff exactly when $\mathcal{R}(\lambda)$ is nonintegral. In that case $A_N \uparrow \mathcal{R}(\lambda)$ and its gap is eventually bounded below by $\lfloor \mathcal{R}(\lambda) \rfloor + 1 - \mathcal{R}(\lambda) > 0$. For an integral remainder the eventual gap is instead the omitted tail itself, smaller than the tested upper bound. Increasing N changes neither M nor denominator coverage. The long paper gives the full endpoint argument in §5.1 and further numerical examples in its Appendix B.6.

The limit of short truncations. As $D \rightarrow \infty$, the comparison (17) cannot hold at cutoffs $N = D + O(1)$ for vectors with $M > 0$ and $V_2 = \dots = V_D = 0$. Indeed, the compulsory divisibility $L_D \mid M$ and (12) give

$$\frac{2|M|}{(N+1)! - 1} \rightarrow \infty,$$

whereas the right side of (17) is at most one. Thus the comparison requires larger cutoffs. This is a support restriction only when the cutoff must equal the largest supported index; it does not exclude a vector with narrower support evaluated later.

Large denominator valuations alone also leave the gap to the next integer uncontrolled. For a fixed prime q , the reduced fractions $1 - 1/q^e$ have denominator q^e , but their gaps to the next integer are $1/q^e \rightarrow 0$.

7 Computation records

An exact non-unit carry at the prime index 67 already forces [every rational denominator to be at least 67](#). The exact GMP carry certificate covers $3 \leq m \leq 300000$. Its unit carries occur at

$$52, 591, 1030, 1407, 1438, 2164, 4258, 10991, 21236.$$

In particular $b_{300000} \neq 1$, giving the exclusion proved in (1). The carry certificate is an exact computation outside Lean; Lean proves the implication from a non-unit carry to the exclusion. The long paper, §6, identifies its program and output, and describes a separate computation through 4000.

The continued-fraction enclosure and its denominator implication are proved in §1. The exact computation gives $N = 7054$, enclosure width 7053 at scale 2^{80000} , and 23449 common partial quotients. It is also outside Lean; the long paper, §6, links its program and output. These two finite calculations are distinct from an unbounded family of non-unit carries or successful remainder comparisons.

A A criterion using the factorial digits of $S - e + 2$

The termwise identity

$$\frac{1}{n! - 1} = \frac{1}{n!} + \frac{1}{n!(n! - 1)}$$

separates the familiar factorial series for $e - 2$ from the number

$$C = \sum_{n \geq 2} \frac{1}{n!(n! - 1)}.$$

Absolute convergence gives

$$C + (e - 2) = S. \tag{18}$$

For $m \geq 2$, define the canonical factorial digit

$$d_m(C) = \lfloor m!C \rfloor - m \lfloor (m-1)!C \rfloor, \quad 0 \leq d_m(C) < m.$$

For $m \geq 3$, the congruence in the next theorem says exactly that $d_m(C) = m - 2$.

The floor convention excludes the alternative expansion with an eventually maximal tail. For example,

$$\frac{1}{2} = \frac{1}{2!} = \sum_{m \geq 3} \frac{m-1}{m!},$$

but the canonical digits of $1/2$ are $d_2 = 1$ and $d_m = 0$ for $m \geq 3$. With this convention, Cantor's factorial expansion [3] represents a rational number exactly when its digits eventually vanish. For $C = S - e + 2$, the following criterion instead requires eventual digit $m - 2$: adding back $e - 2$ supplies the missing $1/m!$ at each index.

Theorem A.1 (rationality and factorial residues). *The following statements are equivalent:*

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1. $S \in \mathbb{Q}$;
2. $(\lfloor m!C \rfloor + 2) \bmod m = 0$ for every sufficiently large m .

Consequently,

$$S \notin \mathbb{Q} \iff (\forall B)(\exists m > B) (\lfloor m!C \rfloor + 2) \bmod m \neq 0. \quad (19)$$

Proof. First suppose $S = a/q$ with $q > 0$, and take m so large that $q \mid (m-1)!$. Write

$$E_m = m! \sum_{2 \leq n \leq m} \frac{1}{n!}.$$

Then E_m is an integer and the omitted scaled tail satisfies $0 < m! \sum_{n > m} 1/n! < 1$. Also, $m!S$ is an integer divisible by m . Multiplying (18) by $m!$ therefore gives

$$\lfloor m!C \rfloor = m!S - E_m - 1.$$

Every summand of E_m except the endpoint $m!/m! = 1$ is divisible by m . Thus $E_m \equiv 1 \pmod{m}$ and $\lfloor m!C \rfloor \equiv -2 \pmod{m}$.

Conversely, assume the displayed congruence from some index onward. Since $0 \leq d_m(C) < m$, for $m \geq 3$ it is equivalent to

$$d_m(C) = m - 2.$$

Choose N beyond the exceptional indices. The canonical factorial expansion of C then has the form

$$C = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C)}{m!} + \sum_{m > N} \frac{m-2}{m!}.$$

Adding $e - 2 = \sum_{m \geq 2} 1/m!$ and using the telescoping identity

$$\sum_{m > N} \frac{m-1}{m!} = \sum_{m > N} \left(\frac{1}{(m-1)!} - \frac{1}{m!} \right) = \frac{1}{N!}$$

gives

$$S = \lfloor C \rfloor + \sum_{m=2}^N \frac{d_m(C) + 1}{m!} + \frac{1}{N!},$$

which is rational. Negating the eventual statement yields the cofinal formulation above. $\square$

Remark. It remains to prove that the displayed congruence fails at arbitrarily large indices for this particular C . Any one finite list of failures, however long, is not enough. The theorem is an equivalence, not an estimate for the distribution of these residues.

A.1 When subtracting the tail changes a floor

The digit uses C , but the carry uses a partial sum. Their floors can differ when the omitted tail crosses an integer. Write $\{x\} = x - \lfloor x \rfloor$ for the fractional part and $\mathbf{1}$ for an indicator, and put

$$\delta_m = m! \sum_{n>m} \frac{1}{n!(n!-1)}, \quad \sigma_m = \mathbf{1}_{\{m!C\} < \delta_m}.$$

Since $0 < \delta_m < 1/((m+1)!-1) < 1$ by comparison with $m! \sum_{n>m} 1/n! < 1$, subtracting this tail lowers a floor by exactly σ_m . Hence

$$Z_m = m! \sum_{n=2}^m \frac{1}{n!} + \lfloor m!C \rfloor + 1 - \sigma_m.$$

The first term is an integer, and its difference from m times the preceding one is 1. Substitution in the definition of b_m therefore gives, for $m \geq 3$,

$$b_m = m - 1 - d_m(C) + \sigma_m - m\sigma_{m-1}. \quad (20)$$

Equality $\{m!C\} = \delta_m$ leaves an integral prefix and gives no correction. A fractional part smaller than the tail crosses an integer, however small the tail is; neither indicator can simply be omitted.

B Guide to the formal sources

The proofs above can be read without following the Lean links. The numbered results carry their Lean proofs in the margin; the links below identify further Lean statements used in the proofs, at fixed earlier revisions of the sources. The carry certificate and the continued-fraction enclosure of Section 7 are exact integer computations outside Lean.

Further formal sources. The Lean sources also give [the explicit vector \$U_9\$](#) of (3) with [its weighted sums](#), and state the congruence (4) as [divisibility of \$V_d\(\lambda\) - M\(\lambda\)\$ by \$d! - 1\$](#) . The identities for adjacent coefficient differences, the recursion that isolates one weighted sum, the vanishing moment, the individual values V_d , and the factor 12 have formal proofs in [the Lean file on the integer basis](#). The basis expansion, the attainable-moment formula and the construction on an arithmetic progression are also written out in [an account of the integer basis](#) and, for the construction, in [an account of the primitive vector](#). The Lean proof of Theorem 2.2 ([the finite formula for the gcd](#)) also checks the scalar recurrence, the equal-block divisibility and the explicit coefficient at twice a prime used in its proof. Checks for the finite gcd calculation are in [the finite gcd verification script](#). The divisibility $\text{lcm}(1, \dots, n) \mid u_n$ and the minimum moment 1380 at $D = 4$ have ordinary proofs, supplemented by exact checks in [a script](#). [Generic gcd lemmas in the Lean sources](#) assume that divisibility; they do not prove it for the factorial coefficients.

The common-denominator bound. The growth $\liminf$ (12), with the least common multiple of $n! - 1$ over $2 \leq n \leq N$ as the common denominator, is the ordinary asymptotic consequence of the finite terminal-block inequality, and Lean checks it as [the \$\liminf\$ bound for the common denominator](#). The finite-block inequality has a separate Lean source, [the inequality for a terminal block](#). The long paper proves the same bound in §2, where the margin links give its Lean proof and an independent Comparator check. This bound concerns a common denominator of the partial sums. It is not a lower bound for the denominator of the series under a rationality assumption.

Links to individual statements. The links below are grouped by the calculation they support.

Successive partial sums and carries: [the condition for two consecutive unit carries](#); [the two denominator-reduction factors telescope](#); [the denominator after two steps](#); [the offset equals the later numerator times the reduction factors](#); [denominator at least the carry index](#); [denominator at least 67](#); [criterion using the next integer](#); [cofinal non-unit carries imply irrationality](#); [the interval condition at the lower endpoint](#).

Selected factors and determinant remainders: [comparison using the factors 1 and a selected prime](#); [the remainder equals a determinant times \$S\$, up to an integer](#).

Finite factorial-weighted sums: [the summand \$W_{n,m}/\(n! - 1\)\$](#) ; [the weight \$W_{n,m}\$](#) ; [the finite sum \$\sum_{n=2}^m W_{n,m}/\(n! - 1\)\$](#) ; [the divisor sum \$\sum_{2 \leq n \leq m, n|m} W_{n,m}\$](#) ; [the factorial recurrence](#); [the quotient when the divisor does not divide the index](#); [the quotient when the divisor divides the index](#); [the term when the divisor does not divide the index](#); [the term when the divisor divides the index](#); [the term at its own index](#); [the coefficient at its own index](#); [the recurrence for this finite sum](#).

Coefficient identities: [weights are integers](#); [the integer factorial weights](#); [the denominator-times-weight identity](#); [the weighted sum](#); [the factorial-weighted moment](#); [the change in a weighted sum at one index](#); [adjacent differences affect only divisor weighted sums](#).

Constant values of the floor in the weights: [the factorisation for \$k \geq 0\$](#) ; [its zero-moment consequence](#); [the case \$d \leq n < 2d\$](#) ; [the support conclusion](#).

The two-term vector: [the vector \$2e_3 - e_4\$](#) ; [the coefficient at index three](#); [the coefficient at index four](#); [the vanishing second weighted sum](#); [the moment of the two-term example](#); [the third weighted sum of the example](#); [the fourth weighted sum of the example](#); [the later weighted sums of the example](#); [the remainder expressed in terms of the tail](#); [the strict rational bounds for the example remainder](#).

Conditional support bounds: [a cubic lower bound for the support parameter](#); [exclusion of the stated eventual cubic upper bound](#); [the support parameter is not little-o of the cubic scale](#); [the finite lower bound with constant three halves](#); [exclusion of an eventual upper bound with constant three halves](#).

Related denominator estimates. Shared prime-power divisibility was used to control the spacing of factorial indices by Erdős and Stewart [13, §3, pp. 516–517]. The subtraction in the proof of (12) is the case $P = -1$ of Luca and Shparlinski's polynomial-shift argument [9, proof of Lemma 5, p. 811], also used by Lai [8, proof of Lemma 2.4, display (2.5)]. These sources supply that subtraction, not the lcm

asymptotic stated here. The extension to a fixed nonzero polynomial shift $n! + P(n)$ uses an eventual cutoff for positivity and nonvanishing [8, Lemma 2.1], attributed there to Luca and Shparlinski [9]. Its proof, including the uniform choice of cutoff and the lower-order gcd loss, is in the long paper, §2. The non-polynomial shift $n! + 2^n - 1$ uses a different, three-index elimination [14, Lemmas 2.1–2.3]; it is not an instance of that extension. Spacing methods for factorial congruences [16, Lemma 2] and the fixed-value multiplicity bound [17, arXiv v1, Theorem 12, p. 16] are relevant comparisons, not ingredients of this proof.

Comparison with denominator savings in p -adic linear forms. A related denominator-saving argument occurs in work of Lai, Lupu and Sprang on linear forms in p -adic zeta values. Here $p \geq 5$ is a fixed prime and n indexes the forms. They save a product Φ_n of prime powers from coefficient-denominator bounds built from $\text{lcm}(1, \dots, n)$ [10, §5]. Its growth rate [10, Lemma 7.3] enters their inequality [10, (8.1)] and irrationality criterion [10, Lemma 2.1, p. 3], quoted from Lai [11, Lemma 2.1, p. 4]. On the same unbounded subsequence, nonzero forms with integer coefficients must have their p -adic absolute values, multiplied by the largest ordinary absolute value of a coefficient, tending to zero. For H_M , the analogous saving is $L_M / \text{den}(H_M)$, whose required asymptotic size is not established here. This is a comparison of denominator control; no p -adic theorem is applied to S .

Factorial-series comparisons. Galambos treats the rationality of Cantor series in [4, Ch. II, §2.1, pp. 21–22]. Koepf and Schmiersau prove irrationality for nonterminating factorial expansions whose digits are not eventually maximal [5, Example 3.2, p. 121]. Appendix A specifies the canonical convention and proves the criterion for $C = S - e + 2$ directly.

The argument in §1 identifies the next integer above a scaled partial sum. This differs from the factorial-tail integrality lemma of Hančl and Tijdeman [6, Lemma 2.1 and the following remark, p. 385]: $m!H_m$ need not be integral; under the denominator-divisibility hypothesis, it is $m!S$ that is integral. The exclusion can also be expressed by saying that the least positive r with $q \mid r!$ is at least 300000. Sondow uses this factorial index for e [15, §3, Theorem 1]; his approximation bound is not applied to S .

For integer-coefficient factorial series, Hančl and Tijdeman prove integrality of normalised rational tails [6, Lemma 2.1 and the following remark, p. 385], and classify the polynomial-coefficient case [6, Theorem 3.1 and Corollary 3.1, pp. 390–391]. The long paper, Appendix B.2, explains the direct application after rewriting S as a factorial series with integer carry-defect coefficients. These results do not remove the indicators in (20).

Selected-prime and mean comparisons. The test with an additional selected prime adds $(2p + 1)C_p q < 2p^2(2p - 1)!$ for a prime $q \mid R_p$ to (14). The long paper, §4, constructs blocks with $q \mid R_p$ and proves the equivalence with the two inequalities. This is an additional restriction, not an equivalent criterion for irrationality. Its §7 gives a sufficient joint mean bound for (15), on the same sets of parameters for both summands; separate estimates on unrelated sets do not suffice.

Additional proofs and comparison files. A related cofactor construction gives a remainder equal to a determinant times S , up to an integer. For a positive starting parameter, its primitive normalization is the progression vector of §3.2, up to sign; the long record, §7, proves this identification. Nonintegrality along the family with least support index tending to infinity is equivalent to irrationality, not a condition established by the construction.

The interval condition at the lower endpoint and the test for $p^2 \mid Z_{2p}$ combine quantities at the same index. Non-unit carries at arbitrarily large $2p$, with p an odd prime, already suffice without that test. These comparisons, the condition for two consecutive unit carries, and an example distinguishing the integer recurrence from its defining floor relation are given in the long paper. It also explains why any fixed collection of denominator factors eventually divides the factorial being removed. The long paper also gives the two simultaneous inequalities for a selected prime, the shared-factor bound for the selected factors, and the eventual absorption of fixed denominator indices.

The unconditional correction (20) is checked in the exact floor correction for the companion series, and the residue congruence for A_M in §4 and the two displayed reciprocal equalities are checked in the prime-power noncancellation criterion. The indices and cofactor residues in §4 come from a separate exact modular scan.

Acknowledgements

The author thanks Wouter van Doorn for advice on exposition: explaining notation when it first appears, avoiding private terminology, and saying how restrictive a conditional hypothesis is. His advice concerned the writing of another note; he has not reviewed the mathematics of this paper. An AI research pass supplied by Will Cook derived the depth-four dual congruence and shorter vectors from the factorial-channel definitions and general moment-ideal theory developed earlier here. OpenAI Codex checked and integrated them; a separate AI pass reviewed the proof.

References

- [1] Paul Erdős. On the irrationality of certain series: problems and results. In Alan Baker (ed.), *New Advances in Transcendence Theory*, Cambridge University Press (1988), pp. 102–109.
- [2] Thomas F. Bloom. Erdős Problem #68. Online resource (2026). Historical access: 28 July 2026; present-page status not reverified.
- [3] Georg Cantor. Über die einfachen Zahlensysteme. *Zeitschrift für Mathematik und Physik* **14** (1869), 121–128.
- [4] János Galambos. *Representations of Real Numbers by Infinite Series*. Lecture Notes in Mathematics 502, Springer (1976).
- [5] Wolfram Koepf and Dieter Schmersau. Irrationality of certain infinite series II. *Analysis* **31** (2011), 117–124.

- [6] Jaroslav Hančl and Robert Tijdeman. [On the irrationality of factorial series](#). *Acta Arithmetica* **118** (4) (2005), 383–401.
- [7] Joel Louwsma and Joseph Martino. [Rational numbers with odd greedy expansion of fixed length](#). Preprint (2023). arXiv:2309.07280.
- [8] Li Lai. [On the largest prime divisor of \$n! + 1\$](#) . *Bulletin of the Australian Mathematical Society* **113** (3) (2026), 390–403. arXiv:2103.14894.
- [9] Florian Luca and Igor E. Shparlinski. [Prime divisors of shifted factorials](#). *Bulletin of the London Mathematical Society* **37** (6) (2005), 809–817.
- [10] Li Lai, Cezar Lupu and Johannes Sprang. [On the irrationality of certain \$p\$ -adic zeta values](#). *Research in the Mathematical Sciences* **12** (4) (2025), article 77. arXiv:2505.23088.
- [11] Li Lai. [On the irrationality of certain 2-adic zeta values](#). *International Journal of Number Theory* **21** (1) (2025), 207–235. arXiv:2304.00816.
- [12] NIST Digital Library of Mathematical Functions. [Continued fractions: convergents](#). Online resource (2026).
- [13] Paul Erdős and Cameron L. Stewart. [On the greatest and least prime factors of \$n! + 1\$](#) . *Journal of the London Mathematical Society* **13** (3) (1976), 513–519.
- [14] Florian Luca and Igor E. Shparlinski. [On the largest prime factor of \$n! + 2^n - 1\$](#) . *Journal de Théorie des Nombres de Bordeaux* **17** (3) (2005), 859–870.
- [15] Jonathan Sondow. [A geometric proof that \$e\$ is irrational and a new measure of its irrationality](#). *American Mathematical Monthly* **113** (7) (2006), 637–641. arXiv:0704.1282. Addendum: *Amer. Math. Monthly* **114** (2007), 659; reading copy v2.
- [16] Cameron L. Stewart. [On the greatest and least prime factors of \$n! + 1\$, II](#). *Publicationes Mathematicae Debrecen* **65** (3–4) (2004), 461–480.
- [17] Moubariz Z. Garaev, Florian Luca and Igor E. Shparlinski. [Character sums and congruences with \$n!\$](#) . *Transactions of the American Mathematical Society* **356** (12) (2004), 5089–5102. arXiv:math/0403422.